

Games and Decisions Under Projection: Blackwell Wedges, Budget Flips, and Prospect Signatures from Declared Consumers

Andrew H. Bond*

San José State University, San José, California, USA

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An information set is the fiber of a projection, many hidden states behind one observed signal, and Blackwell’s ordering of experiments is the decision-theoretic twin of the data-processing inequality. This paper asks whether the behavioral structures of imperfect-information decision problems, the value of information, the signature distortions of prospect theory, reference dependence, and budget-limited equilibrium play, are consumer-relative phenomena, derivable from a declared channel, task, environment, and budget with nothing inserted. Seven experiment runs in exact finite models answer with measurements. A validated instrument layer computes exact values of experiments and certified garbling relations. In an ensemble of two thousand random experiment pairs, 281 pairs were ordered unanimously by all seven declared divergences while certified Blackwell-incomparable, and every one of the 281 carried a task that strictly prefers the divergence-dominated experiment. A constructed pair makes the wedge extreme, one experiment beats the other on every divergence by at least 0.429 while a declared screening task prefers the loser, and the pair is incomparable by proof in both directions. In a zero-sum game where information is bought under a cell budget, the read that maximizes mutual information is worth exactly the no-information value while the task-optimal read at the same budget attains the full-information value, a flip of exactly three fifths, and a strict flip appears in 193 of 200 random games. Budgeted Bayes-optimal readers with no weighting function, no reference point, and no loss-aversion coefficient among their inputs produce regressive linear weighting in closed form, the full inverse-S weighting curve, a crossover that tracks the declared environment, a loss-side response ratio of 1.208 from a loss-heavy environment, and diminishing sensitivity, with every distortion shrinking along the declared budget ladders and vanishing at zero noise. Reference dependence appears exactly where a declared reference functional interacts with an environment-adapted encoder, with an observer spread of 29.6, a ranking reversal with margins 12.4 and 10.8, and a translation-covariance breach of 6.44 that the adapted observer avoids exactly. The rational-inattention logit of Matejka and McKay is replicated to fourteen digits, consideration sets form exactly as the budget prices actions out, and in a coordination game of two budgeted consumers equilibrium attention is complementary, a better-informed opponent makes attention more valuable. Two runs failed their declared bars on their own declarations’ bookkeeping, the records carry the failures, and corrected redeclarations were run afterward. Nothing in this paper is a claim about human beings.

I. INTRODUCTION

This campaign models observation as consumption. A consumer reads a hidden state through a declared channel, spends a declared budget on the read, and acts on a declared task. Earlier stages of the campaign measured what such consumers can distinguish in quantum models [12] and what entropy they observe in reversible classical models [13]. Imperfect-information decision theory has worked with the same objects under other names for more than seventy years. An information set is the fiber of a projection, many hidden histories collapsed behind one observed signal. Blackwell’s ordering of experiments, under which one information source is better than another exactly when it is better for every decision problem [1], is the decision-theoretic twin of the data-processing inequality [9]. Rational inattention prices the read in information units [4, 5]. Prospect theory catalogs the distortions that appear when valuations are measured [2, 3].

The question of this track is whether the behavioral structures of imperfect-information domains are consumer-relative, derivable from declared channel, task, environment, and budget structure with nothing inserted. The distortions of prospect theory are usually presented as departures from rationality. The efficient-coding account holds instead that they are what optimal reading through a limited channel looks like [6, 7]. That account makes an exactly checkable claim in a finite model. Declare the channel, the environment, and the budget, forbid every behavioral ingredient as an input, compute the optimal read, and see which signatures appear.

Three claim types run through the paper and are never conflated. Instrument claims state that values of experiments, garbling relations, and optimal budgeted reads can be computed exactly in finite models, and they are checked against closed forms. Structural claims state that declared consumer structure produces a named behavioral signature without that signature being inserted, or measurably fails to, and they carry the campaign’s exploratory label because they are measured in declared models. Replication claims state that a published theoretical result is reproduced exactly in a declared finite

* andrew.bond@sjsu.edu; ORCID: 0009-0003-2599-6158

instance. Nothing in this paper is a claim about human beings, about empirical behavioral data, or about the rationality of any person, and no run is labeled as supporting such a claim.

An anti-circularity contract governs every run. No probability weighting function appears anywhere in a model's declarations. No reference point is chosen after inspecting outcomes. No loss-aversion coefficient enters as an input. No task list is selected after seeing which tasks favor a wanted verdict. Budgets, channels, priors, and utilities are declared before the runs that bear claims, and any behavioral signature must appear in the measured optimal reads or the claim fails.

The paper reports seven experiment runs and one instrument layer. The instrument layer computes exact values of experiments and certified garbling relations and must pass closed-form controls before anything else runs. The first experiment measures the wedge between scalar fidelity orderings and task orderings and locates it exactly in the region Blackwell's theorem leaves open. The second transplants the campaign's budget flip, task-optimized reads beating fidelity-optimized reads at scarce budget [12], into a game against an adversary. The third asks whether prospect theory's signature distortions emerge from budgeted readers that contain none of them. The fourth locates reference dependence in the choice of a declared reference functional read through an environment-adapted encoder. The fifth replicates the rational-inattention logit exactly and then measures which equilibrium structures survive when every player is a budgeted consumer. Two of the runs failed their declared bars. In both cases every other declared item passed and the failure was an error in the declaration's own bookkeeping, a threshold applied to the wrong object in one case and a double count in the other. The records carry the failed verdicts, the corrected declarations were rerun, and both accounts are reported in full.

II. THE INSTRUMENT LAYER PASSES EXACT CONTROLS

An experiment in Blackwell's sense is a row-stochastic likelihood matrix A with A_{sy} the probability of signal y in state s . For a prior π and a task, a utility matrix $u(a, s)$ over actions and states, the value of the experiment is the expected payoff of the posterior-optimal action,

$$V(A) = \sum_y \max_a \sum_s \pi_s u(a, s) A_{sy}, \quad (1)$$

an exact finite sum. Experiment B is a garbling of A when $B = AM$ for some row-stochastic M , and Blackwell's theorem states that this holds exactly when A is worth at least as much as B for every prior and every task [1]. The instrument layer implements Eq. (1) directly and implements a garbling checker as a projected-gradient search for M whose checked object is the residual $\max |AM - B|$

against a declared feasibility tolerance of 10^{-7} . The solver is only a search. The residual is the evidence.

Five controls with known answers gate the layer, and no later run in the track executes until all five pass. The uninformative and perfect experiments reproduce the no-information and full-information closed forms within the declared 10^{-12} . Across a declared battery of twenty tasks, a battery being a list of tasks fixed before the run, garbling never raises any value, with a worst measured gain of 1.1×10^{-16} . A declared garble is certified feasible at residual 5.6×10^{-17} while its reverse direction is refused at residual 0.260, and a signal permutation is certified in both directions. The binary symmetric channel family is ordered by its error rate, and the checker reproduces that ordering across the declared grid of error rates from 0.05 to 0.45. The last control constructs an incomparable pair, refused in both directions, whose two declared tasks disagree, one prefers the first experiment by 0.05 and the other prefers the second by 0.15. That pair seeds the next section.

III. UNANIMOUS FIDELITY ORDERINGS GUARANTEE NOTHING OUTSIDE COMPARABILITY

The campaign's quantum track found task-optimized encoders beating fidelity-optimized encoders at scarce budget [12]. Blackwell's theorem gives that finding a sharp decision-theoretic frame. A garbling loses for every task, while any scalar fidelity measure is a single number, so scalar orderings can only disagree with task orderings where no garbling relation holds in either direction. This experiment measures the disagreement exactly. The setting is a binary state with uniform prior and experiments with three signals, so every scalar fidelity in the declared battery is an f -divergence between the two conditional signal rows. The battery holds seven, Kullback-Leibler in both directions, total variation, squared Hellinger, chi-squared in both directions, and Jensen-Shannon. A wedge is a pair where all seven divergences strictly order one experiment above the other while some declared task strictly prefers the other.

The declared exhibit is a rare decisive signal. The two experiments are

$$A = \begin{pmatrix} 0.90 & 0.05 & 0.05 \\ 0.05 & 0.05 & 0.90 \end{pmatrix}, \quad B = \begin{pmatrix} 0.98 & 0.019 & 0.001 \\ 0.881 & 0.02 & 0.099 \end{pmatrix}, \quad (2)$$

and the exhibit task is a screening decision, a safe action worth zero in both states against a treat action worth -20 in the healthy state and $+1$ in the sick state, so treating pays only above the posterior threshold $20/21$. A is far more informative on every divergence, with margins from 0.429 to 15.1 across the battery, but its best posterior is 0.947 and never clears the threshold, so its value for this task is exactly zero. B is nearly uninformative except for one rare signal, probability 0.05, whose posterior is 0.99,

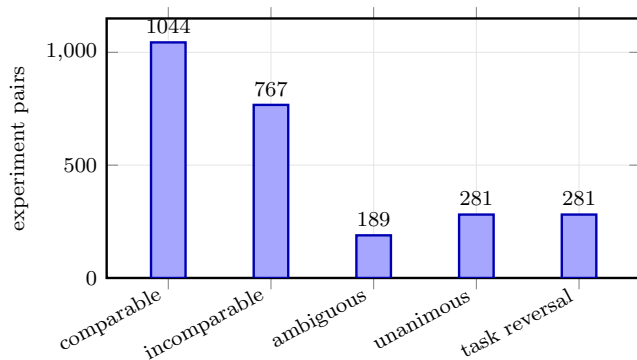


FIG. 1. The declared ensemble of 2000 random experiment pairs. The first three bars partition the ensemble by garbling certificate, comparable when a certificate is feasible in some direction, incomparable when both residuals exceed the declared 0.01, ambiguous otherwise and excluded as declared. The last two bars are subsets of the incomparable class, pairs ordered unanimously by all seven divergences and pairs whose unanimous ordering is reversed by a task in the declared battery. The two subsets are equal at 281. Every count is read from the committed evidence record.

and its value is 0.0395. The task prefers B by 0.0395 while every declared fidelity prefers A by at least 0.429.

Three further items were declared with the exhibit. A data-processing control drew 500 pairs related by a declared random garble and checked them against the battery of 205 tasks, 200 random three-action utilities and five screening thresholds, all seed-fixed. No divergence rose under any garble, the smallest margin in the required direction measured 1.8×10^{-4} , and no task gained more than 3.3×10^{-16} , which is Blackwell's theorem plus the data-processing inequality holding in the instrument, and a violation would have indicted the instrument rather than the theorems. A localization item classified a declared ensemble of 2000 random pairs by garbling certificate and required every wedge to fall in the incomparable class. Figure 1 shows the outcome, 1044 comparable, 767 incomparable, 189 ambiguous, and zero wedges among comparable pairs, with 144 wedges among the excluded ambiguous pairs. The prevalence measurement was declared with no bar. Among the 767 certified-incomparable pairs, 281 were ordered unanimously by all seven divergences, and every one of the 281 carried a task reversal in the declared battery, with a maximum reversal margin of 0.0776 and a median of 0.0214. In this ensemble, unanimous scalar-fidelity ordering failed as a task guarantee in every instance outside certified comparability.

The recorded verdict of this run is FAIL. The exhibit's declared bar required both garbling-search residuals to exceed 0.01, a round number declared without computing the scale it would be compared against. The reverse residual measured 0.4198 and passed. The forward residual measured 0.004486 and failed the bar, while every divergence margin and the task margin passed. The verdict stands as recorded, computed from the declared items.

The redeclared run replaced the threshold with a proof and passed all five of its items. Incomparability of the exhibit is established in both directions without any residual bar. In one direction, the columns of any garbling of A are conic combinations of A 's columns, so a target column (t_0, t_1) with $t_0 < r t_1$, where r is the smallest first-to-second-row ratio among A 's columns, forces a residual of at least $(r t_1 - t_0)/(1 + r)$ for every row-stochastic M . Computed from the declared matrices, the bound is 0.004263 against its declared bar of 10^{-3} , more than four orders of magnitude above the feasibility tolerance, so B is not a garbling of A . In the other direction, total variation is an f -divergence, a garbling cannot raise it, and A 's total variation exceeds B 's by 0.751, so A is not a garbling of B . The consistency item reads the failed bar's own number against the bound. The search residual 0.004486 sits 5.2 percent above the analytic bound, so the search was measuring genuine infeasibility, and the original failure was a bar thresholding a search residual instead of certifying infeasibility. The exhibit therefore stands proved, every declared f -divergence orders A above B by at least 0.429, the declared screening task strictly prefers B , and the pair is Blackwell-incomparable in both directions. Together with the ensemble measurement, the wedge that the campaign's budget flip exploited is exhibited exactly in decision form and localized to exactly the region Blackwell's theorem leaves open.

IV. THE FIDELITY-OPTIMAL PURCHASE BUYS NOTHING AGAINST AN ADVERSARY

The second experiment transplants the budget flip into a game. Six hidden states carry a declared prior. Player one buys a deterministic coarsening of the state, a partition into at most k cells for budget k , and the choice of partition is public. Player two observes nothing. Both then move simultaneously in a zero-sum game with state-dependent payoffs, player one maximizing. With two actions for player two the value is exact. Player one's best response to a mixture q is greedy within each cell, so the game value is the minimum over q of a convex piecewise-linear function, minimized exactly by enumerating the crossing points of the per-cell action lines. All 203 partitions of six states are enumerated. The fidelity-optimal read at budget k maximizes mutual information between state and cell, which for a deterministic coarsening is the entropy of the cell masses, with ties collected within 10^{-12} and the flip always charged against the best-valued member of the tie set. The task-optimal read maximizes game value. The flip at a budget is the task-optimal value minus the best fidelity-optimal value.

The declared exhibit is a stake game built so that the flip has rational closed forms. States are pairs of a payoff-relevant bit with prior mass 0.8 and 0.2 and a payoff-irrelevant trit uniform within each bit value, so the six priors are 4/15 three times and 1/15 three times. Player one guesses the bit, earning +1 on a match and -2

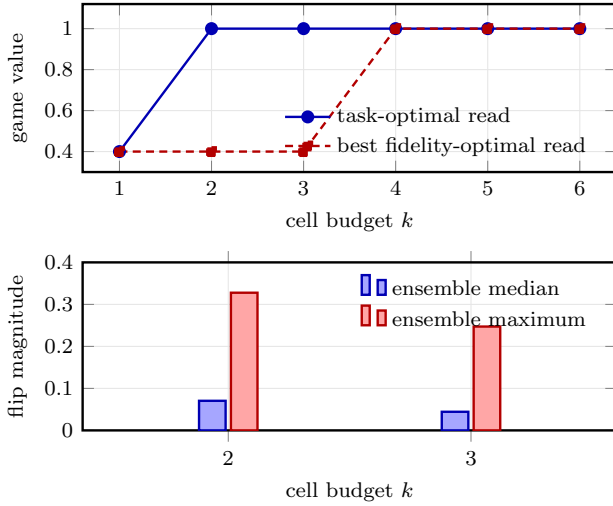


FIG. 2. The budget flip in the stake game and in the ensemble. The upper panel shows the exhibit’s value ladder. At budgets 2 and 3 the task-optimal read attains the full-information value 1 while every fidelity-optimal read is worth 0.4, the no-information value, and the two selection rules agree at large budgets. The lower panel shows the ensemble flip magnitudes over 200 random games, median over flipped games and maximum, at budgets 2 and 3. A strict flip occurred in 193 of 200 games at budget 2 and 188 of 200 at budget 3. Every number is read from the committed evidence records.

on a miss, multiplied by a stake of 1 or 3 chosen by player two. The declared closed forms follow from the structure. The no-information value is $2/5$. The task-optimal two-cell read is the bit split, worth exactly 1, which is also the full-information value. The fidelity-optimal two-cell reads are the mass-balancing partitions that split $8/15$ against $7/15$, each mixing the bit inside one cell, each worth exactly $2/5$. The fidelity-optimal spend of the budget buys exactly nothing, and the flip is exactly $3/5$.

The first run of this experiment recorded verdict FAIL, and the failure is an integer in the declaration. Every measured value matched its declared closed form within 10^{-15} , the task-optimal two-cell value measured exactly 1, every fidelity-optimal two-cell read measured 0.4, the flip measured 0.6 against its bar of 0.5, the ladders were monotone, and no partition beat the finest, which is the zero-sum theorem that more information never hurts, carried as an instrument-indicting control. The one failed item was the declared count of tied fidelity-optimal two-cell partitions, six declared against three measured. The declaration had tallied each two-cell partition once from each of its cells. The mass- $7/15$ subsets are exactly the complements of the mass- $8/15$ subsets, three partitions, not six. The redeclared run corrected that count, held every other object fixed, and passed all six items.

The ensemble measurement, declared with no bar and identical across both runs, shows the flip is generic rather than constructed. Over 200 random games with Dirichlet priors, three player-one actions, two player-two actions, and payoffs uniform on $[-1, 1]$, all seed-fixed, the task-

optimal read strictly beat every fidelity-optimal read in 193 of 200 games at budget 2, with median flip 0.070 and maximum 0.328, and in 188 of 200 games at budget 3. Figure 2 shows the exhibit ladder and the ensemble magnitudes. The campaign’s flip is now measured in a game against an adversary. At scarce budget, buying the read that maximizes mutual information is generically the wrong purchase.

V. PROSPECT SIGNATURES EMERGE FROM CONSUMERS THAT CONTAIN NONE OF THEM

Prospect theory’s weighting function overweights small probabilities, underweights large ones, crosses the identity near one third to one half, and is steep at both ends [2, 3]. Its value function is asymmetric between gains and losses and exhibits diminishing sensitivity. The efficient-coding account predicts that these shapes are properties of optimal reading through a limited channel under an environmental prior [6, 7]. This experiment declares three budgeted consumers, none containing a weighting function, a reference point chosen after outcomes, or a loss-aversion coefficient, and measures which signatures their optimal reads exhibit. The verdict is computed from audits on the machinery alone. The findings carry declared directional bars and are recorded pass or fail individually, and either outcome is a result.

The first consumer reads a win probability p through n Bernoulli draws under a $\text{Beta}(2, 2)$ environmental prior and reports the posterior mean, with the sample budget n on a declared ladder from 1 to 200. Its implied weighting has the closed form

$$w(p) = \frac{2 + np}{4 + n}, \quad (3)$$

and the measured read matches Eq. (3) within 6.8×10^{-14} at every n with maximum curvature 1.5×10^{-14} . This consumer produces overweighting of small probabilities and underweighting of large ones, regression toward the prior mean, with no inverse-S curvature at all. The weighting is exactly linear.

The second consumer is a log-odds reader. The hidden state is the log-odds ℓ of the probability, on a grid from -6 to 6 in steps of 0.01 , with a Gaussian environmental prior of variance 1.5 . The channel adds discrete Gaussian noise of width σ to the coded value, and the read is the exact posterior mean of $p = 1/(1 + e^{-\ell})$. The implied weight $w(p)$ is the channel average of the read. The declared budget ladder is σ equal to 1.0 , 0.5 , 0.25 , and exactly 0 as the identity control. Figure 3 shows the measured curve at $\sigma = 1.0$. It is the full inverse-S, from nothing but a log-odds code, a Gaussian budget, and Bayes-optimal reading. The crossover is exactly 0.5 by symmetry. The endpoint secant slopes measure 3.00 at both ends against a declared bar of 1.1 , the middle secant measures 0.51 against a bar of 0.9 , and the overweighting and underweighting margins on the declared bands measure 0.099 where 0.01 was

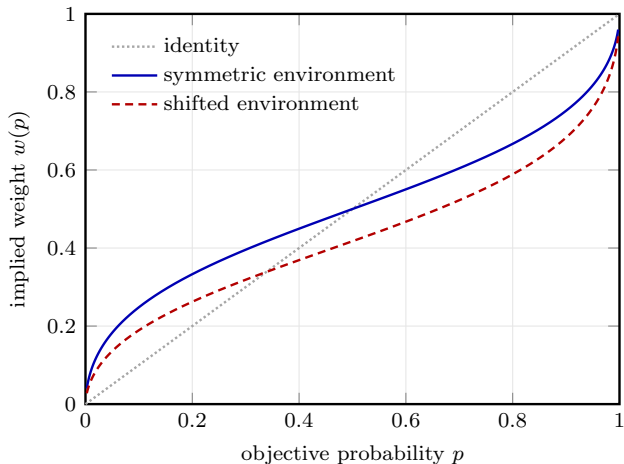


FIG. 3. The implied probability weighting of the log-odds reader at budget sigma 1.0. Under the symmetric environment the curve is the full inverse-S, crossing the identity at exactly 0.5, overweighting below and underweighting above, with endpoint secant slopes 3.00 and middle slope 0.51. Under the declared shifted environment, centered at log-odds minus one, the crossover moves to 0.338. The curve is read from a committed figure-data artifact recomputed by the committed GD-3 instrument and bound by hash to the evidence record.

required. The declared shifted environment, the same reader with the prior centered at log-odds -1 , moves the crossover to 0.338, inside its declared bar of 0.35. The distortion tracks the environment, not the machinery.

The third consumer reads outcome magnitudes from -100 to 100 through the compressive code $\text{sign}(x) \ln(1 + |x|)$ with Gaussian channel noise, under a symmetric Laplace environment and, in a declared variant, an asymmetric environment whose loss side is heavier, scale 40 against 20, declared on the efficient-coding rationale that the question is whether a budgeted read transmits environmental asymmetry into valuation asymmetry. Under the symmetric environment the read is odd to 1.5×10^{-13} , so the machinery inserts no asymmetry. Under the loss-heavy environment the measured response ratio at ± 50 is 1.208, the loss side read less compressed, with nothing resembling a loss-aversion coefficient anywhere in the model. Diminishing sensitivity is unambiguous, the declared coarse second differences on the gain side are all negative with maximum -0.092 and mean -0.402 .

The audits that compute the verdict all passed. The first consumer matches its closed form. The zero-noise consumers read the identity within 10^{-12} , so the machinery inserts nothing. Symmetric environments produce exactly symmetric reads. And the maximum deviation from the identity strictly decreases along every declared budget ladder, from 0.392 to 0.0096 for the sample reader, from 0.149 to zero for the log-odds reader, and from 45.0 to zero for the magnitude reader, so every distortion is budget-borne and vanishes when the budget constraint does. All five declared findings passed their directional

TABLE I. Valuations of the six declared gambles by the four declared observers at budget sigma 2.5, rounded to two decimals from the committed evidence record. The mean column is the gamble's expected value, which every observer reproduces at zero noise within 1.8×10^{-15} . B6 is B5 shifted by ten. The adapted observer's B6 entry equals its B5 entry plus ten exactly, while the status-quo observer misses that shift by 6.44.

Gamble	mean	status quo	adapted	lower med.	lagged
B1	5	0.63	5.00	-24.56	5.00
B2	5	3.41	5.00	6.16	5.00
B3	10	18.02	10.00	10.00	10.00
B4	0	0.00	0.00	0.00	0.00
B5	0	-11.80	-11.80	-13.71	0.00
B6	10	-8.24	-1.80	-3.71	10.00

bars. The signature set of prospect theory, regressive weighting, the inverse-S, environment-borne loss asymmetry, and diminishing sensitivity, emerged in declared budgeted consumers with none of it among the inputs.

VI. REFERENCE DEPENDENCE IS THE DECLARED FUNCTIONAL READ THROUGH AN ADAPTED ENCODER

A design note recorded in the protocol shaped this model and is itself a finding. A posterior-mean read decoded under the gamble's own distribution is exactly unbiased in expectation, iterated expectation cancels every reference effect, so reference dependence cannot live there. The declared model is therefore an encoder adapted to a fixed environment and presented with gambles it was not adapted to, which is also the efficient-coding picture, and the reference point is the encoder's adaptation level.

The reader encodes deviations $d = x - R$ from a reference R through the compressive code $\text{sign}(d) \ln(1 + |d|)$, a discrete Gaussian channel of width σ acts on the code, and the read is the exact posterior mean of d under the declared environmental prior, a symmetric Laplace of scale 30. The valuation of a gamble is the mean reference plus the expected read of its deviations. Four reference functionals are declared, none chosen after outcomes. The status-quo observer takes $R = 0$. The adapted observer takes R equal to the gamble's mean, the expectation-based reference of Köszegi and Rabin in its simplest form [8]. The third observer takes the lower median. The lagged observer takes the previous outcome of an independent two-period presentation. A declared battery of six gambles probes them, including a pair related by a shift of ten.

The audits all passed. At zero noise every valuation equals the gamble mean within 1.8×10^{-15} . The symmetry anchors hold within 7.1×10^{-14} at every budget, the adapted observer reads the symmetric gambles at their means, and the lagged observer reads every gamble at its mean because independent two-period differences are

TABLE II. The rational-inattention ladder of the declared instance. Columns are the information cost, the unconditional probability of the safe action, the expected payoff of the optimal strategy, and the information it buys in nats. The safe action is priced out of the consideration set at every cost up to 1 and carries everything at cost 5. All numbers are from the committed evidence record.

λ	safe prob.	value	info (nats)
10^{-6}	0	1.000000	0.693147
0.05	3.1×10^{-23}	1.000000	0.693147
0.2	7.4×10^{-16}	0.993307	0.652968
1.0	4.8×10^{-13}	0.731059	0.110944
5.0	1.000000	0.600000	3.3×10^{-15}

symmetric. The adapted observer’s translation covariance is exact, its valuation of the shifted gamble equals its valuation of the base gamble plus ten with measured deviation zero against the declared 10^{-10} tolerance. And the maximum deviation of any valuation from the gamble mean falls strictly along the budget ladder, 29.6 at $\sigma = 2.5$, 20.1 at 1.0, machine zero at 0.

Table I shows the valuations at the scarcest budget. The maximum spread across observers on a single gamble is 29.6, fifty-nine times its declared bar. A ranking reversal sits inside the declared battery, the status-quo observer prefers B1 to B5 by 12.4 while the lower-median observer prefers B5 to B1 by 10.8. The status-quo observer breaks translation covariance by 6.44 on the declared shift while the adapted observer holds it exactly, so which reference effects appear is decided by the declared functional, and every appearance is predicted by the detector model. Reference dependence in this model is not a preference anomaly. It is the declared functional’s interaction with an environment-tuned budgeted encoder, with every difference between observers derived from declared structure.

VII. THE RATIONAL-INATTENTION LOGIT REPLICATES EXACTLY AND EQUILIBRIUM ATTENTION IS COMPLEMENTARY

The last experiment has a replication half and a campaign half. The replication target is the result of Matejka and McKay that a rational-inattention consumer, maximizing expected payoff minus λ times the mutual information between state and action, chooses by a weighted logit [4, 5]. The declared instance has four uniform states and three actions, a safe action worth 0.6 everywhere, and two risky actions each worth 1 on their own half of the states. The optimum is computed by Blahut-Arimoto iteration [10, 11] with the logit step in log space, run to a fixed-point residual of 10^{-14} , over the declared cost ladder λ from 10^{-6} to 5.

Five declared bars gate the replication and all five passed. Stationarity, the unconditional action distribution equal to the marginal of the conditional strategy, holds

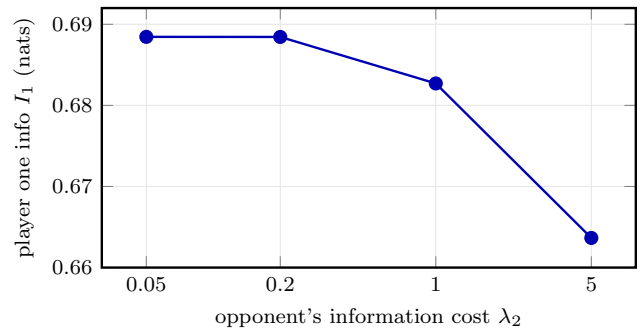


FIG. 4. Attention complementarity in the coordination game. Player one’s information cost is fixed at 0.2 while the opponent’s cost rises along the declared ladder. Player one’s equilibrium information falls from 0.6884 to 0.6637 nats as the opponent coarsens, so a better-informed opponent makes attention more valuable. All four points are read from the committed evidence record.

with residual exactly zero. The weighted-logit identity of Matejka and McKay,

$$P(a | s) = \frac{p_0(a) e^{u(a,s)/\lambda}}{\sum_b p_0(b) e^{u(b,s)/\lambda}}, \quad (4)$$

with p_0 the unconditional action distribution, holds at the fixed point to 9.5×10^{-15} . None of 200 declared random feasible perturbations of the conditional strategy improves the net objective, the recorded maximum improvement is exactly zero. The limits land on their closed forms, value 1 with information equal to $\ln 2$ to the last digit at vanishing cost, and prior play at cost 5, where the safe action carries unconditional probability 1 within 10^{-9} . Table II shows the ladder. The consideration-set phenomenon appears exactly as the replication target describes. The safe action is excluded at every cost up to 1, with unconditional probability 3×10^{-23} at cost 0.05, and at cost 1 the optimum is interior, value 0.7311 with information 0.1109 for a net objective of 0.6201, strictly better than both pure safe play at 0.6 and full attention. Feasible actions drop from the consideration set exactly as the budget prices them out.

The campaign half puts two budgeted consumers in a coordination game. Two uniform states, two actions each, payoff 1 for matching the state plus 0.5 for matching the other player, so given the opponent’s conditional strategy Q the effective payoff is $u(a, s) = \mathbf{1}[a=s] + 0.5 Q(a | s)$ and best responses are Blahut-Arimoto solves, iterated from uniform play to a joint residual of 10^{-12} . All three declared equilibrium bars passed. The iteration converges at every declared budget pair. Symmetric budgets give the symmetric equilibrium, including state-flip symmetry, with deviation 2.1×10^{-14} . Along the common budget ladder the equilibrium information is nonincreasing, matching accuracy at the finest budget is above 0.95, and at every ladder point the measured equilibrium accuracy

q sits on the independent closed-form curve

$$q = \frac{1}{1 + e^{-(1+0.5(2q-1))/\lambda}}, \quad (5)$$

the scalar fixed point of the weighted logit applied to the coordination payoff, with residual 2.1×10^{-14} . The Shannon-cost solution is interior at every finite budget, and the measured equilibrium lands on the curve that an entirely separate route computes.

The declared finding measures how one player’s budget moves the other player’s equilibrium attention. Player one’s cost is fixed at 0.2 while the opponent’s cost ladders over 0.05, 0.2, 1, and 5. Figure 4 shows the result. Player one’s equilibrium information falls monotonically from 0.6884 to 0.6637 nats as the opponent’s budget coarsens, passing its declared direction and minimum-drop bars. In this game attention is complementary. Information bought by one player raises the value of information to the other, exactly as declared before the run.

VIII. WHAT THIS PAPER ESTABLISHES

Table III registers the runs with their recorded verdicts. Read together they answer the track’s question inside its declared scope. The value of information is task-relative in a precise sense, scalar fidelity orderings carry no task guarantee outside certified Blackwell comparability, and in the measured ensemble the unanimous-fidelity wedge was never empty of task reversals. The budget flip survives an adversary, and the fidelity-optimal purchase can buy exactly nothing while the task-optimal purchase at the same budget buys everything. The signature distortions of prospect theory, regressive weighting, the inverse-S, loss-side asymmetry, and diminishing sensitivity, are producible by declared budgeted consumers that contain none of them, with each distortion shrinking as the budget grows and vanishing at zero noise, and with the shape tracking the declared environment. Reference dependence in these models is the choice of a declared reference functional read through an environment-adapted encoder, with every observer difference predicted by the detector model. The rational-inattention logit is repli-

cated exactly, and in the one equilibrium measured, attention is complementary across players.

Two verdicts in the ledger are FAIL, and both failures were located in the declarations rather than in the measured structures, a residual threshold where a proof was needed and a double-counted tie set. The corrected redeclarations passed with unchanged measured values.

These are structural results about declared finite models carrying the exploratory label of the campaign’s evidence-grading scheme. They show that the named behavioral signatures can arise from channel, task, environment, and budget structure alone. They do not show that human behavior arises this way, and no experiment in this track can bear on that question.

IX. METHODS AND REPRODUCIBILITY

All runs follow the campaign’s standing discipline. Priors, utilities, channels, budgets, batteries, seeds, tolerances, and bars were declared in the track document before the runs they govern, per the anti-circularity contract, and verdicts are computed from the declared items, never written first. Every result-bearing run writes an append-only JSON evidence record containing the declared parameters, the measured outcomes, and a canonical SHA-256 hash of the record body. The records for this paper are `gd0-instrument.json`, `gd1-flip-blackwell.json`, `gd1b-exhibit-certificate.json`, `gd2-budget-flip-game.json`, `gd2b-budget-flip-game.json`, `gd3-prospect-signatures.json`, `gd4-reference-functionals.json`, and `gd5-budgeted-equilibrium.json` in the campaign repository, with the committed runners alongside them. The failed GD-1 and GD-2 records are retained unchanged, and the corrected declarations are separate records, so the sequence of verdicts is itself part of the evidence. Every number and every figure coordinate in this paper is read from those committed records. The figure tables are generated deterministically by a committed script with no free parameters, and the one plotted curve not stored in a verdict record, the weighting curve of Fig. 3, is a committed figure-data artifact, `gd3-figdata-curve.json`, recomputed by the committed GD-3 instrument and bound by hash to the record it accompanies. A reducer may summarize records but may not rewrite them.

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TABLE III. The ledger of the games-and-decisions track. Verdicts are recorded exactly as computed from the declared items of each run, including the two failures. Measured in model, exploratory label, means an exact measurement in a declared finite construction with append-only evidence records and pre-declared bars, seeds, and batteries. Key numbers are quoted from the committed records.

Run	Verdict	Key measurements	Label
GD-0 instrument layer	validated	closed-form control values matched within 10^{-12} , garbling never raised any of 20 task values (worst gain 1.1×10^{-16}), declared garble certified at residual 5.6×10^{-17} against reverse 0.260, binary-channel ordering reproduced, incomparable seed pair with task margins 0.05 and 0.15	instrument
GD-1 wedge ensemble	FAIL	exhibit failed its residual-threshold bar, 0.0045 against the declared 0.01, while the data-processing control (500 garbles, battery of 205) and localization (zero wedges among 1044 comparable pairs) passed, and the ensemble measured 767 incomparable pairs, 281 unanimous, 281 of 281 with task reversals (max 0.0776, median 0.0214)	measured in model, exploratory
GD-1b exhibit certificate	PASS	incomparability proved both ways, conic bound 0.004263 against bar 10^{-3} , total-variation margin 0.751, all seven divergence margins at least 0.429, task margin 0.0395, search residual 5.2 percent above the analytic bound	measured in model, exploratory
GD-2 budget flip in a game	FAIL	every closed form matched within 10^{-15} and the flip matched its declared 3/5, but the declaration counted six tied fidelity-optimal partitions where the instrument counted three, a double count from both cells of each two-cell partition	measured in model, exploratory
GD-2b corrected declaration	PASS	task-optimal two-cell value exactly 1, every fidelity-optimal two-cell read at the no-information value 0.4 within 10^{-15} , flip at the declared 3/5, ensemble flip in 193 of 200 games at budget 2 (median 0.070, max 0.328) and 188 of 200 at budget 3	measured in model, exploratory
GD-3 prospect signatures	PASS	all eight audits and all five findings, linear regressive weighting to 6.8×10^{-14} , inverse-S with secants 3.00 and 0.51 and crossover 0.5 moving to 0.338 under the shifted environment, loss ratio 1.208 with no loss-aversion input, second differences max -0.092	measured in model, exploratory
GD-4 reference functionals	PASS	audits exact (zero-noise identity 1.8×10^{-15} , adapted covariance exact), observer spread 29.6, ranking reversal 12.4 against 10.8, status-quo covariance breach 6.44	measured in model, exploratory
GD-5 logit and equilibrium	PASS	weighted-logit identity 9.5×10^{-15} , stationarity 0, 200 perturbations never improve, limits exact (value 1, information $\ln 2$, prior play at cost 5), consideration-set exclusion 3×10^{-23} , equilibrium on the scalar fixed-point curve to 2.1×10^{-14} , complementarity 0.6884 to 0.6637	replication, and measured in model, exploratory
Scope	non-claim	nothing in this track is evidence about human decision-making, about prospect theory as psychology, or about the rationality of any person	non-claim

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